

Fazul Al Rehman

Offensive Security Researcher | Bug Bounty Hunter

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EDUCATION

BS Computer Science Institute of Business Administration (IBA), Karachi <i>Final Year Project: IntelliScore — AI-Based Credit Scoring System (in collaboration with Meezan Bank)</i>	2021 – 2025
HSSC (Pre-Engineering) Superior College of Sciences	94%, 2019

PROFESSIONAL EXPERIENCE

Independent Security Researcher Full-Time Bug Bounty Hunter <i>Offensive security research and responsible disclosure across enterprise, e-commerce, gaming, and Web3 targets.</i>	Jan 2026 – Present
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Samsung Pass , Critical Production Vulnerability	Critical, Bugcrowd
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- Discovered a leaked **Jenkins** CI/CD **Slack** bot token that bypassed authentication as an authorized internal integration, granting admin-level access to sensitive operational channels.
- Mapped a significant portion of **Samsung Pass** infrastructure through the exposed access, including live dashboards, **Terraform** configurations, SSH key rotations, and internal environment credentials; responsibly disclosed via **Bugcrowd** for a confirmed \$10k bounty on a platform handling biometric data for millions of devices.

Walmart Marketplace, Payment Access-Control Bypass	Critical
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- Found a critical access-control (IDOR) flaw in **Walmart** Marketplace Seller Central where manipulating request parameters granted unauthorized read access and could reroute seller payments, placing an estimated \$50M in payment flow at risk of drainage.

Large-Scale Credential Exposure via Dorking	Disclosure in progress
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- Used systematic search-engine dorking to surface 20,000+ exposed private keys and API secrets spanning **AWS**, **Stripe**, **Twilio**, **Razorpay**, **OpenAI**, **Anthropic**, **Shopify**, **Twitch**, and **GitHub**.
- Aggregated **GitHub** keys exposed auto-deploying access to 1,600+ private industrial repositories; also surfaced **Shopify** secret keys affecting multiple e-commerce brands (including Shark Tank India–featured startups, ushopmall, and **Greyson Clothiers**). Coordinating responsible disclosure with affected organizations.

EA Origin , Security-Question Authentication Bypass	Auth bypass
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- Identified a security-question bypass in **EA Origin**'s account-recovery flow, enabling authentication without the legitimate answers.

Dell , Invoice-PDF Authorization Bypass	Auth bypass
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- Reported an authorization bypass on **Dell** allowing retrieval of invoice PDFs through serial-number enumeration.

Web3 Security Research	Ongoing
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- Auditing smart contracts affiliated with **Cantina** and **Immunefi** as part of ongoing Web3 security research.

Teaching Assistant Software Architecture & Design	Aug 2025 – Dec 2025
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- Mentored students on software architecture principles and design patterns; supported course delivery and assessment.

Mindsquared Consultant	Jan 2024 – Jun 2024
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- Built and deployed data pipelines driving profitability roadmaps across **Walmart** and **Amazon** marketplace operations.
- Supported 50+ companies in reaching profitable states through automated analytics and execution workflows.

INDEPENDENT SECURITY ASSESSMENT

IBA Karachi , University Portal Security Assessment	Responsibly disclosed
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- Identified multiple critical vulnerabilities across the portals of IBA Karachi, one of Pakistan's top universities, and responsibly disclosed them to administration, prompting a full migration to a new portal now in production use.
- Findings included a database exposure leaking confidential personal records (including stored .png files) and an authentication bypass affecting faculty accounts up to super-admin level.

PROJECTS

Attacking & Defending Confidential VMs , Hypervisor Interrupt Injection on AMD SEV-SNP & Intel TDX	Computer Security, IBA Karachi
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- Led a security-research project on a published class of attacks (HECKLER) in which an untrusted hypervisor abuses its retained control over asynchronous event delivery to inject unexpected interrupts into hardware-isolated Confidential VMs on **AMD SEV-SNP** and **Intel TDX** Trusted Execution Environments — a gap left open even when guest memory and CPU state are fully encrypted.
- Built a research testbed on an **AMD EPYC** (Milan/Genoa) platform across three iterations — from a stock **QEMU**/modified-kernel setup, through a containerized build, to a lean patched-QEMU instance with a custom **Python** control harness over the QEMU Machine Protocol (QMP) — and characterized guest-kernel behavior under injected NMI and Machine-Check exceptions via forensic dmesg

timing analysis.

- Proposed and evaluated **Guest_Defense**, a multi-layered mitigation combining guest-side filtering of deprecated legacy interrupt pathways with context-aware validation that an incoming interrupt matches the CPU's expected state, mitigating malicious state changes at minimal performance overhead; authored under Dr. Faisal Iradat.
- Reverse-engineered production Windows software — **Camtasia Studio**, **VEGAS**, and **7-Zip** — using binary analysis and disassembly to map program structure, control flow, and licensing behavior, building the low-level familiarity that underpins vulnerability and malware analysis.

SHA-256 Across Kernel, User Space & System Calls, Linux Systems Programming

Operating Systems, IBA Karachi

- Built and compiled **three separate custom Linux kernels** in a **WSL** environment to isolate each implementation path, implementing **SHA-256** as a loadable **kernel module** via the kernel crypto API (`crypto/hash.h`) logging to `dmesg`, as a user-space application over **OpenSSL**, and through a **custom system call** exposing kernel cryptographic functionality to user programs.
- Benchmarked hash computation across all three paths on identical inputs, quantifying the **kernel/user-space boundary-crossing overhead** and isolating where syscall and context-switch costs dominate.
- Analyzed the security trade-offs of each environment — privilege boundaries, access control, and the attack surface of running cryptographic code in **ring 0** versus user space — in a detailed technical report.

IntelliScore, AI Credit Scoring Platform for Meezan Bank

Final Year Project, IBA Karachi

Led a four-person team from a one-line industry brief to a deployed, bank-facing AI product.

- Owned product direction end-to-end for **Meezan Bank**, Pakistan's largest Islamic bank, converting an undefined brief into a scoped roadmap: ran requirements discovery across six executive stakeholder sessions, negotiated a data-sharing NDA with the bank's legal and risk-assessment teams, and absorbed three mid-flight scope pivots (consumer loans → MyBykea bike financing → auto financing) without slipping delivery, replacing an ECIB manual review process that took days to clear ten applications.
- Architected a **semi-supervised teacher-student distillation** pipeline over 93,000 anonymized records and 32 features: a research-backed scoring formula labeled a 50k partition, a fine-tuned **MLPRegressor** (128–64–32) distilled that logic to pseudo-label a 30k partition, and a **HistGradientBoostingRegressor** trained across both at 1.0/0.2 differential sample weights — a deliberate call that stopped the model from collapsing into a replica of its own teacher formula.
- Fine-tuned the ensemble from validation MSE 15.67 to **3.12** at $R^2 = 0.9971$, holding $R^2 = 0.99$ on a fully held-out 13,000-record partition at **sub-2-second** inference, proving the model generalized beyond the formula rather than memorizing it.
- Built the platform's conversational layer as a fine-tuned **LLM** chatbot grounded in **RAG**, applying text analysis over applicant profiles and scoring rationale to convert opaque black-box output into an explainable natural-language narrative of why a score was assigned and what moves it.
- Ran the build-versus-buy evaluation on inference infrastructure, benchmarking a self-hosted **Qwen 2.5** RAG stack against the **Gemini** API across cost, latency, and GPU footprint, and shipped the production platform in **Django**, **PostgreSQL**, and **Tailwind CSS** with multi-user auth and a comparative-analytics dashboard.

Nested Learning & Continual Adaptation, LLM Catastrophic-Forgetting Research

Neural Networks & Deep Learning, IBA Karachi

- Reproduced the core mechanisms of **Nested Learning** (2025), a frontier reformulation of deep networks as a hierarchy of nested optimization processes at different time scales that recasts the optimizer itself as a memory system and targets **catastrophic forgetting** in deployed models.
- Implemented the framework from scratch in **PyTorch**: three nested fast/medium/slow modules each driven by its own optimizer and learning rate, a **Continuum Memory System** using EMA shadow weights, LSTM-based context-flow tracking, and a **self-modifying meta-learner** that rewrites its own gradient-update rule at each step — the concrete realization of the paper's in-context self-revision claim.
- Benchmarked the nested architecture against a standard MLP baseline on synthetic regression and classification tasks, visualizing per-timescale gradient flow to empirically confirm that optimization behaves as multi-rate memory rather than a static process.

AI Résumé Generator, Fine-Tuned LLM for Structured Text Generation

Introduction to Text Analytics, IBA Karachi

- Fine-tuned an open-source **LLM (Llama-2 7B)** with parameter-efficient **LoRA** adaptation on a curated corpus of just 500 résumés, engineering the data pipeline and training regime to extract maximum signal from a deliberately small dataset.
- Grounded generation in a **RAG** retrieval layer over the corpus to enforce structure, tone, and domain relevance, producing tailored output that outperformed baseline **ChatGPT** results on the same prompts in head-to-head evaluation.

Parallel Matrix Convolution, Multithreaded Image Filtering

Parallel & Distributed Computing, IBA Karachi

- Parallelized 2D matrix convolution — the core operation behind image filtering — in **C++** with **OpenMP**, decomposing the input into four quadrants processed by 8 concurrent threads to cut execution time over the serial baseline.
- Managed per-thread output aggregation and persisted the input, filter, and result matrices to separate files, handling the data partitioning and recombination that parallel decomposition requires.

Real-Time E-Commerce Analytics Platform, Streaming Big Data Pipeline

Big Data Analytics, IBA Karachi

- Built an end-to-end streaming data platform over 3–4 GB of high-velocity e-commerce event data, ingesting through **Apache Kafka** into a hybrid **Hadoop/HBase** and **Redis** persistence layer and processing in **Apache Spark**, orchestrated as scheduled DAGs in **Apache Airflow** and fully containerized in **Docker** for reproducible deployment across multiple nodes.
- Shipped three distinct analytics workloads against the same live pipeline — **fraud detection**, **customer analytics**, and **demand fore-**

casting — alongside exploratory statistical analysis, surfacing every model through **Python** dashboards served from **Redis**-cached aggregates for low-latency reads.

- Instrumented a separate operator-facing admin dashboard exposing live ingestion throughput and HBase query telemetry, giving the platform genuine end-to-end observability rather than demo-only reporting.

CartNest, Full-Stack Multi-Role E-Commerce Platform

Web Application Development, IBA Karachi

- Built a complete e-commerce platform with a **React** single-page frontend (React Router, Tailwind CSS, Chart.js) served by a **Node.js/Express** REST API over a **PostgreSQL** database, containerized end-to-end with **Docker Compose**.
- Implemented distinct **buyer, seller, and admin** roles with **JWT**-based authentication and **bcrypt** password hashing, covering product catalog and listings, user and seller account management, and image uploads via **Multer**.
- Structured the backend cleanly into routes, controllers, and middleware with a SQL migration and seed workflow (`init.sql`, `migrations.sql`, `seed.sql`), and surfaced analytics through Chart.js dashboards.

Student Forum Portal, Full-Stack Community Platform

Database Systems, IBA Karachi

- Built a full-featured student discussion forum with a **React** (TypeScript) frontend and a **Node.js/TypeScript** backend over a **SQLite** database, delivering threaded posts, comments, user accounts, and content browsing end-to-end.
- Designed a normalized relational schema (users, threads, posts, comments, categories) with enforced **foreign-key constraints** for referential integrity, and indexed frequently queried columns to keep thread and comment loads fast.
- Used **parameterized queries** throughout the data-access layer to eliminate SQL-injection risk, and wrapped multi-step writes in **transactions** to guarantee consistency.

NayaPay Redesign, Human-Computer Interaction

250 screens, Figma

- Redesigned Pakistan's **NayaPay** fintech wallet end-to-end to remove navigational ambiguity from its payment, billing, and onboarding flows; delivered 250 high-fidelity **Figma** screens backed by an 11-component design system (navigation bar, tab bar, biometric prompt, QR scanner).
- Built complete **Urdu** right-to-left localization as a mirrored 124-screen set alongside the 125-screen English build, making the wallet usable for non-English-speaking users rather than translating labels in place.
- Covered 12 end-to-end flows including signup, biometric and passcode login, password recovery, transfers, card management, and bill payment, with accessibility controls for font scaling and light/dark theming as first-class settings screens.

Autonomous Maze-Solving Robot, Arduino & Discrete Logic

3rd of 45 teams

- Built two independent from-scratch implementations of an autonomous maze-solving car driven by three IR sensors through a dual H-bridge motor driver: one in embedded C on **Arduino**, and one as pure combinational hardware with no microcontroller and no firmware.
- Derived the full truth table over the sensor input space and minimized it with **Karnaugh maps** into four Boolean output equations, realizing them in discrete logic ICs with hand-wired interconnects; the same verified equations drive the Arduino firmware unchanged, making the two builds functionally identical by construction.
- Placed 3rd out of 45 competing teams, finishing as the 3rd-fastest car in the maze run.

TECHNICAL SKILLS

Security: Bug bounty, CTF authoring, reverse engineering, binary analysis, QEMU, TEEs (SEV-SNP / TDX), smart-contract auditing, applied cryptography

Languages: Python, JavaScript / TypeScript, C / C++, SQL, Solidity, Embedded C (Arduino)

AI / ML: PyTorch, scikit-learn, neural networks, gradient boosting, teacher-student distillation, LoRA fine-tuning, RAG, LLMs (Llama-2, Qwen 2.5, Gemini API), CUDA C/C++

Data Engineering: Apache Spark, Kafka, Airflow, Hadoop, HDFS, HBase, Hive, MongoDB, Redis, PostgreSQL, SQLite

Web: React, Node.js / Express, Django, REST APIs, JWT auth, Tailwind CSS

Tools: Docker, Git, Linux, Figma, Cisco Packet Tracer

CERTIFICATIONS

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| • NVIDIA Deep Learning Institute — Getting Started with Accelerated Computing in CUDA C/C++ | 2025 |
| • Cisco Networking Academy — Getting Started with Cisco Packet Tracer; Exploring Networking with Cisco Packet Tracer | 2025 |

ACHIEVEMENTS & LEADERSHIP

- **Winner** — **Trading Bazaar**, IBA Karachi.
- **IBA Data Science Society** — Designed and authored the **CTF** challenges for the society's cybersecurity track and served as judge, validating solutions and scoring participants.
- **CryptoHack** — Completed the Introduction, General, Symmetric Ciphers, Diffie-Hellman, and Hash Functions categories on the applied-cryptography platform.